Randomized and low-rank approximation

Critical-point count for corank-one approximation with a fixed zero

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Topic: Structured low-rank approximation and Euclidean distance degree

Rating rationale: The target is an exact formula for a constrained approximation problem in every dimension. It would quantify the algebraic complexity of enumerating candidate nearest singular matrices with a prescribed zero.

Resolution

Solved affirmatively, 11 September 2026. George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA, proves the exact formula $\mathrm{EDdeg}(V_n) = 5n - 7$ for every integer $n \geq 3$. The Theorem and Sections 1-5 give a quadratic spectral curve, an exact stationarity resultant of degree $5n - 7$, and a bijection with the smooth-locus critical points. The argument proves the necessary generic nondegeneracy and reducedness, including a simultaneous exceptional-locus exclusion in every dimension. Complete proof PDF · Standalone XeLaTeX source.

The proof uses the original complex bilinear distance, generic complex data, one fixed entry and the smooth-locus definition. The pre-existing dimension-two exception is retained. All cases of the displayed target are settled; counts of real critical points and other zero patterns are not asserted. Kubjas, Sodomaco and Tsigaridas retain credit for the conjecture and its finite supporting computations.

The complete AI-assisted argument passed a separate independent Codex-agent review, including an independently implemented universal polynomial-identity check. A distinct coordinating-agent audit also passed. These are informal automated reviews, not external human peer review or formal proof-assistant verification. The submission record preserves exact reviewed-source hashes, source checks and the public eligibility audit. The difficulty, importance and rationale above are retained historical ratings of the original question.

Statement

For $n \geq 3$, let

$$V_n = \{X \in \mathbb{C}^{n\times n} : \det X = 0,\ x_{11} = 0\}.$$

For generic $U \in \mathbb{C}^{n\times n}$, consider the bilinear squared distance

$$d_U(X) = \sum_{i,j=1}^{n} (x_{ij} - u_{ij})^2.$$

Its critical points on the smooth locus of V_n satisfy $\mathrm{d}(d_U)_X[Z] = 0$ for every tangent vector $Z \in T_X V_n$. Their finite number for generic U, meaning outside a proper algebraic exceptional set, is the *Euclidean distance degree* $\mathrm{EDdeg}(V_n)$. There is no complex conjugation in this definition; over the reals the objective is squared Frobenius distance.

Conjecture (Kubjas–Sodomaco–Tsigaridas, Conjecture 5.1, nontrivial dimensions).

$$\mathrm{EDdeg}(V_n) = 5n - 7 \qquad \text{for every } n \geq 3.$$

The dimension restriction is explicit here: at $n = 2$, the equations reduce to $x_{11} = 0$ and $x_{12}x_{21} = 0$, a union of two linear spaces with ED degree two. The printed conjecture omits a lower bound on n, while its supporting table begins at three. That exceptional endpoint is not asserted to satisfy the formula.

Evidence and numerical significance

Table 2 of the source reports $8, 13, 18, 23, 28, 33, 38, 43$ for $n = 3, \ldots, 10$, using numerical critical-point computations checked against symbolic ideal degrees. These finite computations support the conjecture but do not establish the formula for arbitrary n.

Truncated SVD solves unconstrained Frobenius approximation, but a fixed zero changes its critical equations. This target concerns corank-one approximation in general square matrices; the symmetric rank-two zero-pattern problem has different geometry and rank constraints.

References and status check

- K. Kubjas, L. Sodomaco and E. Tsigaridas, *Exact solutions in low-rank approximation with zeros*, Linear Algebra and its Applications 641 (2022), 67–97. DOI; current author manuscript, 29 January 2022. Conjecture 5.1 and Table 2, manuscript p.19; §2 defines the critical-point and ED-degree conventions.

On 2026-09-11, checked the current arXiv version, the authors' publication lists, and targeted title, conjecture-number, ED-degree, proof and 2026 searches. No later proof or counterexample for $n \geq 3$ was located. The formulation and table were checked in arXiv v2; a separate publisher PDF was not obtained. The scope correction at $n = 2$ is disclosed above. This is a bounded literature check.